

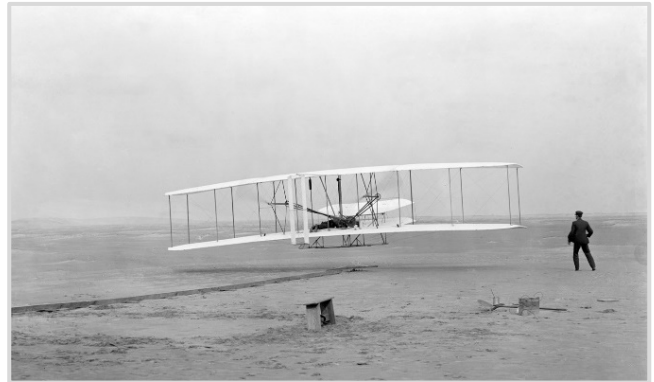
BASICS FOR AIR TRAFFIC CONTROL – PRINCIPLES OF FLIGHT

INTRODUCTION

What forces did Orville and Wilbur Wright have to overcome in order to get this aircraft off the ground?

Knowledge of basic aerodynamics will help you communicate accurately and professionally with pilots concerning their aircraft. This knowledge will be used daily on a routine basis. Occasionally, during an aircraft emergency, an understanding of basic aerodynamics may be invaluable to you as an air traffic control specialist.

The purpose of this module is to provide basic aeronautical information that will help you communicate with pilots concerning the operation of their aircraft.



BASIC AERONAUTICAL INFORMATION

Purpose: The purpose of this lesson is to describe the forces that give an aircraft lift and move it through the air.

Objectives:

- Identify primary and secondary sources of lift
- Identify types and parts of airfoils
- Identify forces affecting flight, their interrelationships, and their effects on aircraft performance

References for this lesson are as follows:

- FAA-H-8083-25, Pilot's Handbook of Aeronautical Knowledge

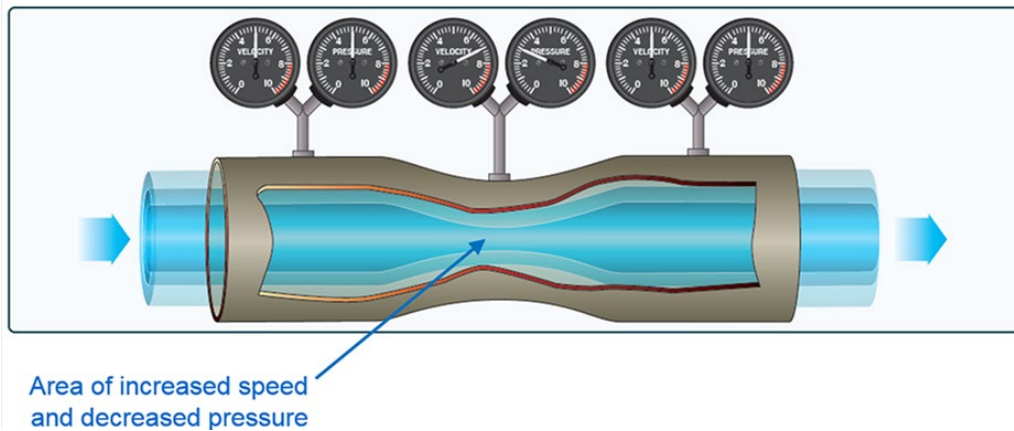
Theories of Flight

To understand what allows an aircraft to fly (how an airplane produces lift), basic knowledge of Bernoulli's Principle and one of Newton's Laws is necessary.

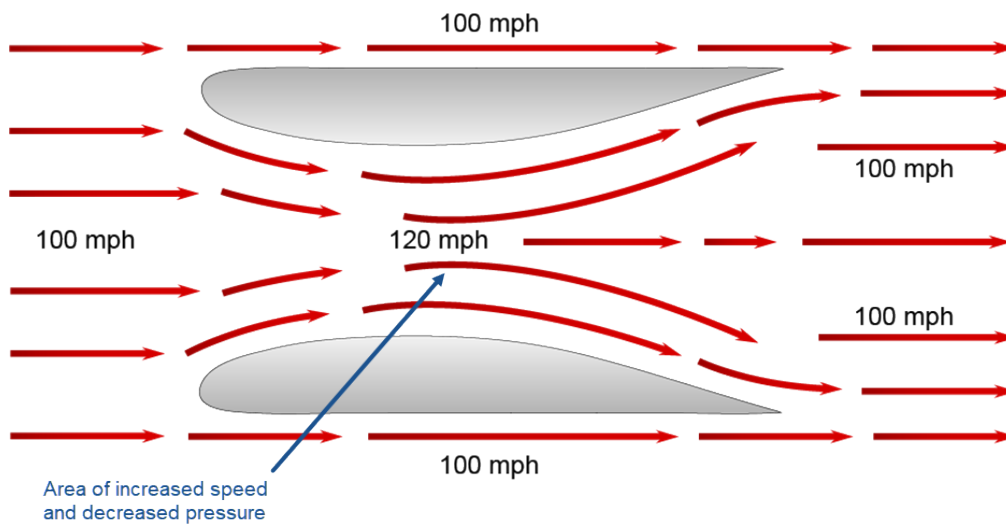
Bernoulli's Principle

Bernoulli's Principle states, in part, that "the internal pressure of a fluid (liquid or gas) decreases at points where the speed of the fluid increases."

- Flow through a tube with a reduced cross sectional area increases fluid speed and decreases fluid pressure

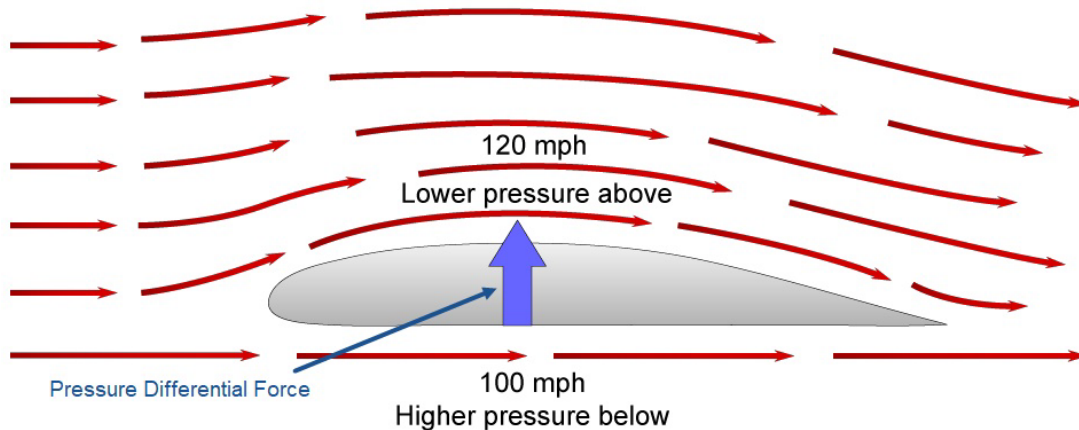


The tube can be replaced by two airfoils and cause the same effects.



The pressure differential around an airfoil is the primary source of lift.

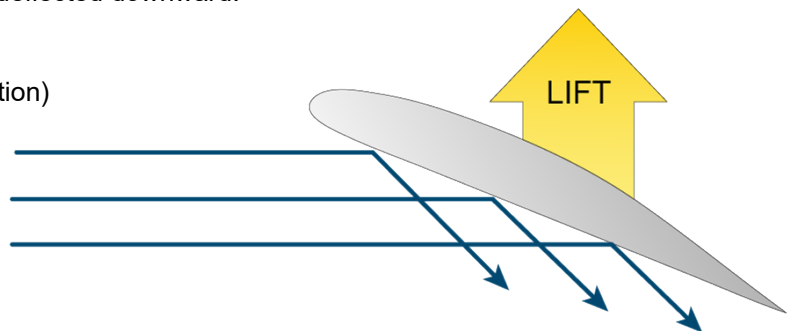
- A pressure differential occurs when there is a **pressure difference** between **opposing sides of a surface**
- A pressure differential causes the higher pressure area below the airfoil to try to equalize pressure by pushing (lifting) the airfoil toward the lower pressure area above
- This lift is the result of Bernoulli's Principle



Newton's Third Law of Motion

For every action, there is **an equal and opposite reaction**. A secondary source of lift is an upward force generated by air striking the underside of an airfoil and being deflected downward.

- In this graphic:
 - Air is striking the underside of an airfoil (action)
 - The wing is being pushed up (reaction)



Knowledge Check A

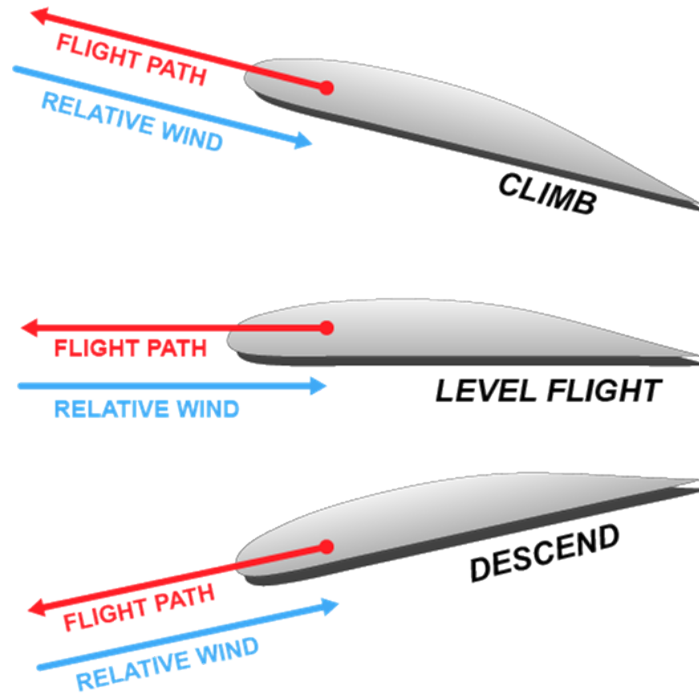
REVIEW what you have learned so far about basic aeronautical information. ANSWER the questions listed below.

1. The primary source of lift on an airfoil is created by a differential in _____. (Select the correct answer.)
 - ☐ Temperature
 - ☒ Pressure
 - ☐ Reaction
2. The statement "the internal pressure of a fluid decreases at points where the speed of the fluid increases" is a part of _____. (Select the correct answer.)
 - ☒ Bernoulli's Principle
 - ☐ Newton's Law of Motion
 - ☐ Hindenburg's Theory

Relative Wind

Relative Wind is the direction of the airflow produced by an object moving through the air.

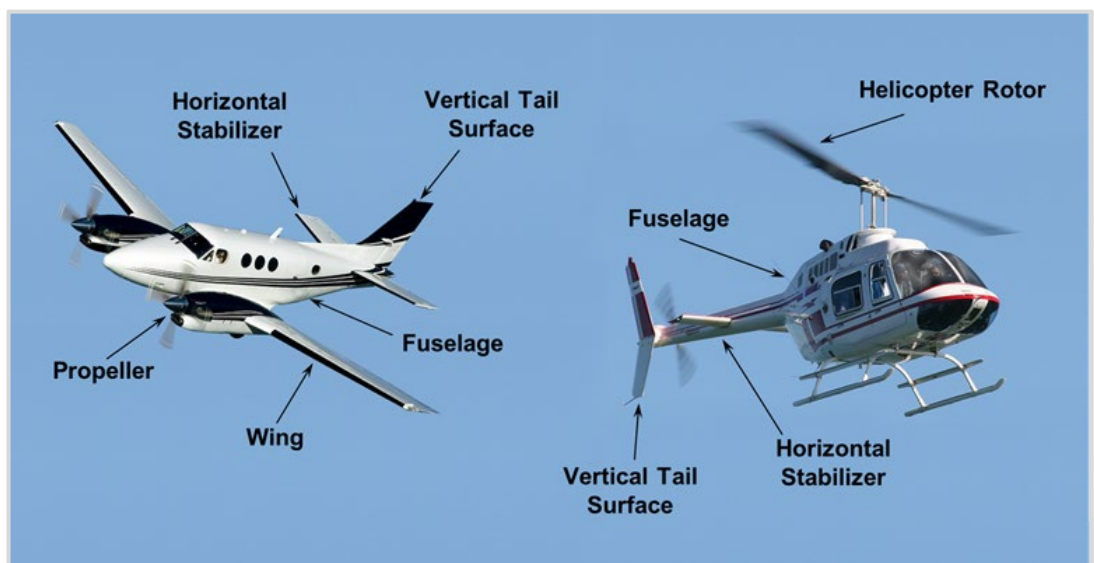
- The relative wind for an aircraft in flight flows in a direction parallel with and opposite to the direction of flight
- The actual flight path of the aircraft determines the direction of the relative wind



Airfoils

Types of airfoils on aircraft are:

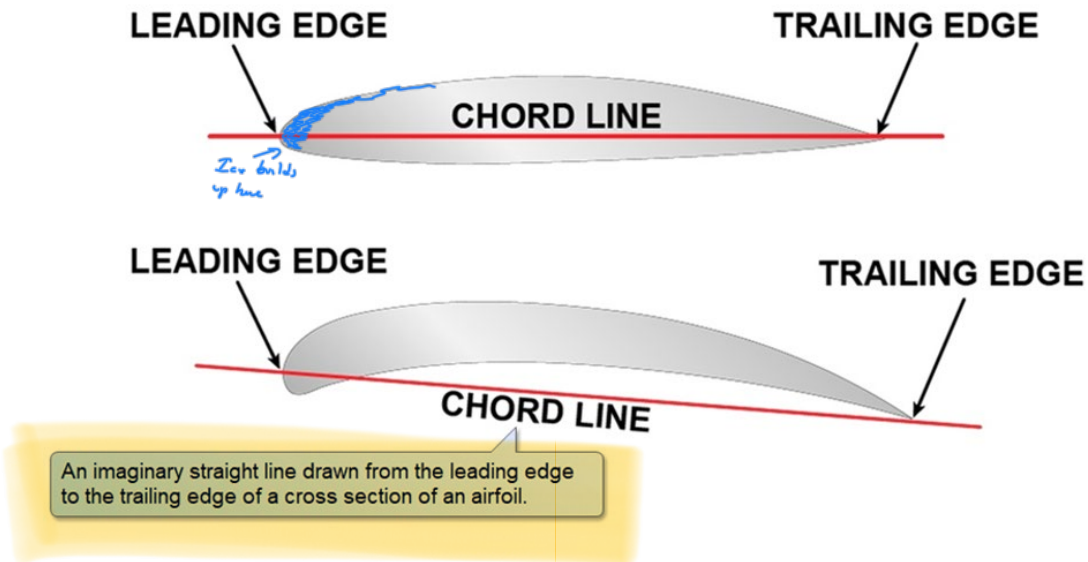
- Wing
- Propeller
- Helicopter rotor
- Horizontal stabilizer
- Vertical tail surfaces
- Fuselage



The three principal airfoils that produce lift on an aircraft are:

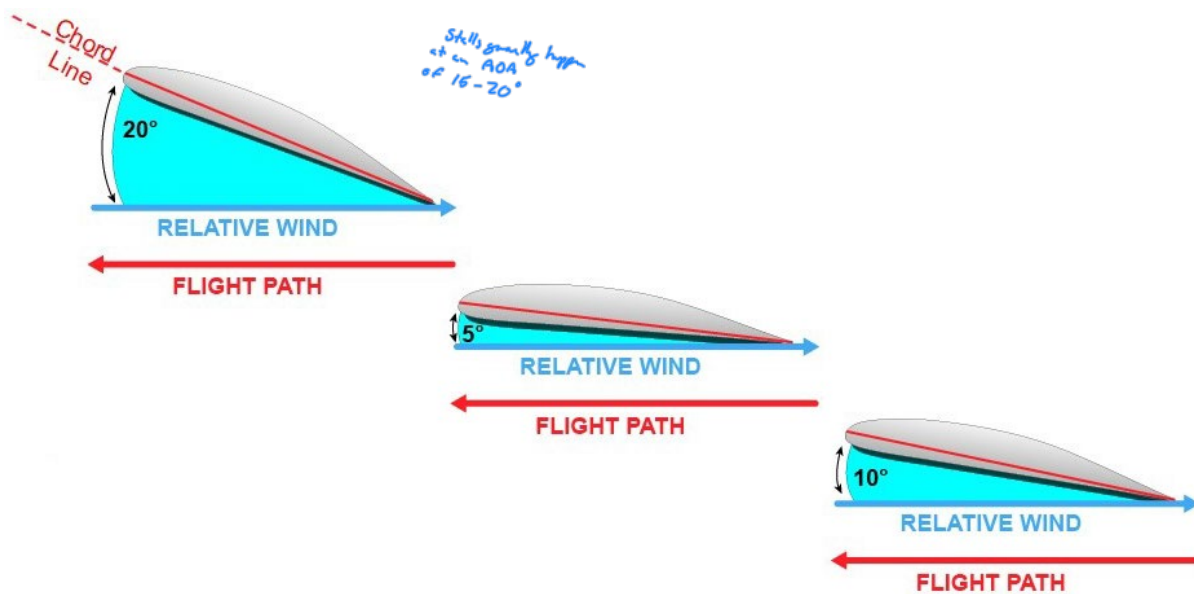
- Wing
- Horizontal tail surfaces
- Propeller (lift produced in a forward direction)

Parts of an Airfoil



Angle of Attack

The **angle of attack** is the angle at which relative wind meets an airfoil. It is the angle that is formed by the chord of the airfoil and the direction of the relative wind.



Note: The angle of attack is based on the relative wind, not the ground.

Camber

The **camber** of an airfoil is the characteristic curve of its upper and lower surfaces. Generally, the upper camber is more pronounced, while the lower camber is comparatively flat. This causes the velocity of the airflow immediately above the wing to be much higher than that below the wing.

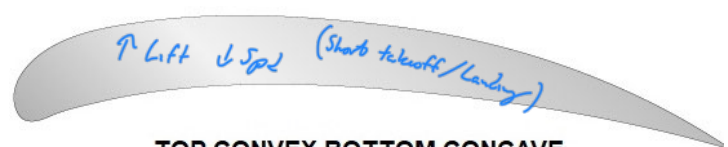
- Lower camber refers to the curvature of the lower surface
- Upper camber refers to the curvature of the upper surface
- The camber or curvature of a wing is designed according to the:
 - Type of aircraft
 - Planned speed of the aircraft
 - Weight of the aircraft
 - Planned use of the aircraft



EQUAL CURVATURE (SYMMETRICAL)



TOP CONVEX BOTTOM FLAT



TOP CONVEX BOTTOM CONCAVE

Wing Planforms

The **wing planform** is the shape or form of a wing as viewed from above. It may be long and tapered, short and rectangular, or various other shapes.

- The planform design is dependent on the use of the aircraft
 - **Examples:** The faster the aircraft, the thinner the airfoil to reduce drag; the thinner the airfoil, the more surface area needed to produce lift
- The amount of lift generated by the wing depends upon several factors:
 - Speed of the wing through the air
 - Angle of attack
 - Planform of the wing
 - Wing area
 - Density of the air
 - Camber



Tapered leading edge,
straight trailing edge



Tapered leading edge
and trailing edge



Delta wing



Sweptback
wings



Straight leading
and trailing edges



Straight leading edge,
tapered trailing edge

✓ Knowledge Check B

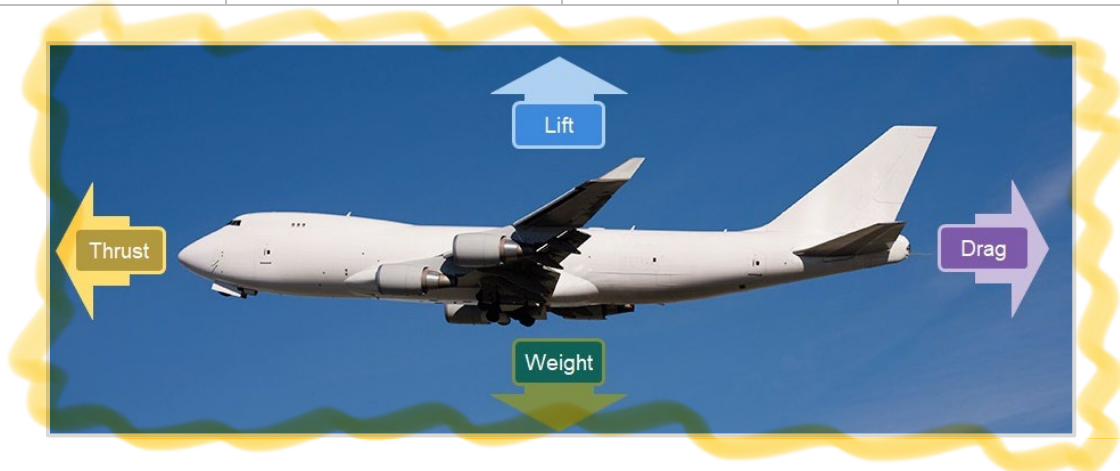
REVIEW what you have learned so far about basic aeronautical information. **ANSWER** the questions listed below.

- What is the curvature of the airfoil from the leading edge to the trailing edge? (Select the correct answer.)
 - ☒ Camber
 - ☐ Equalizer
 - ☐ Airfoil line
- What are the three principal airfoils? (Select all correct answers that apply.)
 - ☒ Wing
 - ☐ Fuselage
 - ☒ Horizontal tail surfaces
 - ☒ Propeller
 - ☐ Tail rudder

Forces Affecting Flight

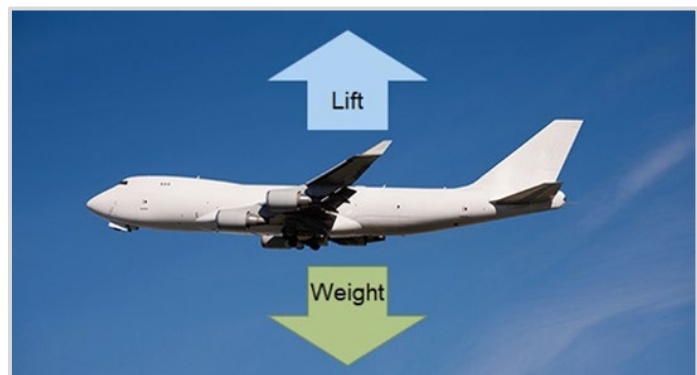
Four Forces Affecting Flight

Lift	Drag	Weight	Thrust
Upward force created by an airfoil when it is moved through the air.	Rearward-acting force that resists the forward movement of the airplane through the air.	Downward force that tends to draw all bodies vertically toward the center of the Earth.	Man-made force that pulls or pushes the aircraft through the air.



Interrelationship of Lift and Weight

- In **straight and level flight (constant altitude)**, lift counterbalances the aircraft's weight, or:
 - When lift and weight are in equilibrium, the aircraft neither gains nor loses altitude
- If lift is greater than weight, the aircraft will climb
- If weight is greater than lift, the aircraft will descend



Interrelationship of Thrust and Drag

- In **straight and level flight**, thrust and drag are equal in magnitude if a constant airspeed is being maintained
- **Thrust is controlled by the throttle**
 - As more throttle is applied, more thrust is produced
- When the thrust of the propeller is increased, thrust momentarily exceeds drag and the airspeed will increase, provided straight and level flight is maintained
 - **With an increase in airspeed, drag increases rapidly**
 - As soon as thrust and drag become equalized, the airspeed will again become constant



Knowledge Check C

REVIEW what you have learned so far about basic aeronautical information. ANSWER the questions listed below.

1. Match the terms with the definitions. Enter your answers in the spaces below.

- | | | | |
|----------|--|----|-----------------|
| <u>B</u> | The angle at which relative wind meets an airfoil. | a. | Relative wind |
| <u>C</u> | The characteristic curve of an airfoil's upper and lower surfaces. | b. | Angle of attack |
| <u>O</u> | The shape or form of a wing as viewed from above. | c. | Camber |
| <u>A</u> | The direction of the airflow produced by an object moving through the air. | d. | Wing planform |

Basic Aeronautical Information Summary

It is hard to imagine how a 735,000-pound aircraft is able to get off the ground and stay airborne. The mechanics of flight are highly complex. Learning the basic flight principles will allow you to more effectively perform your job as an Air Traffic Controller.

EFFECTS OF ATMOSPHERE ON AIRCRAFT PERFORMANCE

Purpose: The purpose of this lesson is to explain how atmospheric conditions affect aircraft performance.

Objective:

- Identify effects of altitude, temperature, and humidity on aircraft performance

References for this lesson are as follows:

- FAA-H-8083-25, Pilot's Handbook of Aeronautical Knowledge
- The Aeronautical Information Manual (AIM)

Atmospheric Properties

The atmosphere is an envelope of air that surrounds the Earth and rests upon its surface. It is as much a part of the Earth as is land and water. The atmosphere is made up of a mixture of gases that reaches almost 350 miles from the surface of the Earth.



Nature of the Atmosphere

Three key properties of the atmosphere that affect air density and aircraft performance are:

- Temperature
- Altitude
- Water vapor (humidity)

3 Hr

Hot

High

Humid

Note: References to aircraft performance in this lesson include length of takeoff roll, initial rate of climb, and length of landing roll.

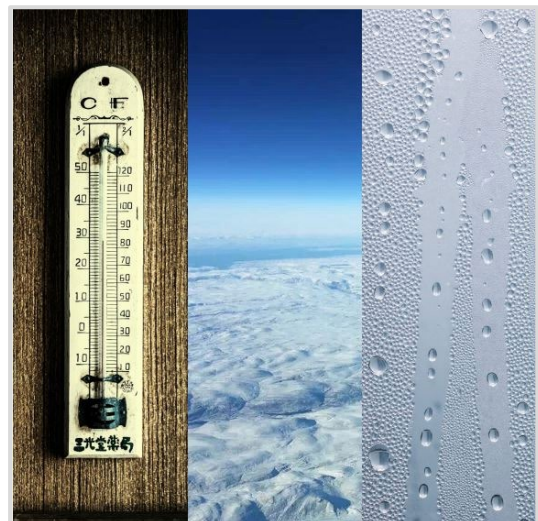
Less dense:

WARM AIR

HIGHER ALTITUDE

LOWER PRESSURE

HIGHER HUMIDITY



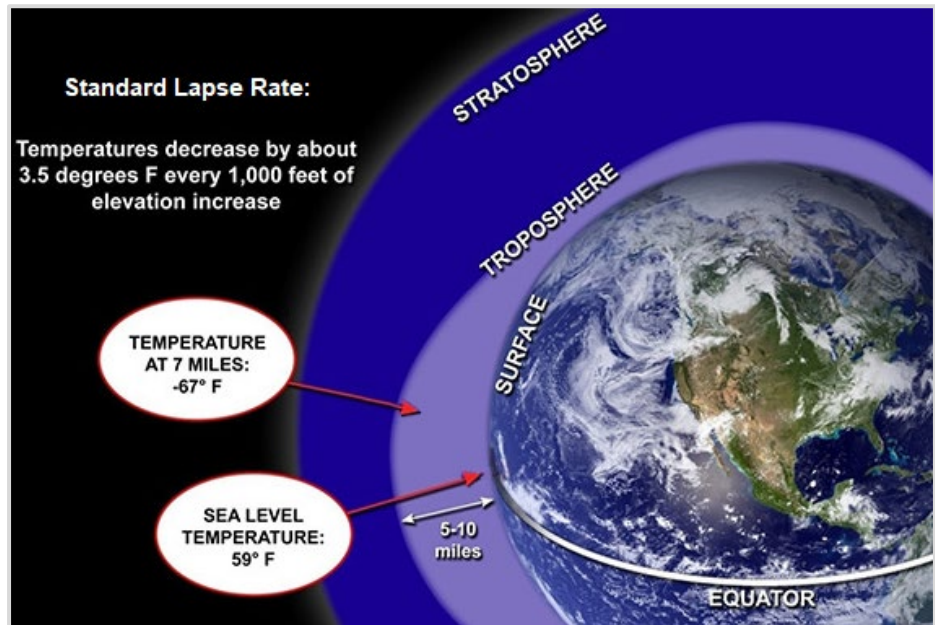
Atmosphere and Temperature

Near the surface, the air is relatively warm from contact with the Earth.

- Surface temperatures change frequently and are relatively warmer during the day and summer and cooler during night and winter
- Due to constantly changing atmospheric conditions, a standard reference was developed
- The standard surface temperature at sea level is 15° Celsius (59° Fahrenheit)

As altitude increases, temperature decreases.

- A decrease of temperature with an increase in altitude is called a lapse rate
- The standard lapse rate is approximately 2°C (3.5° Fahrenheit) per thousand feet
- Cold air is more dense than warm air



Altitude and Pressure

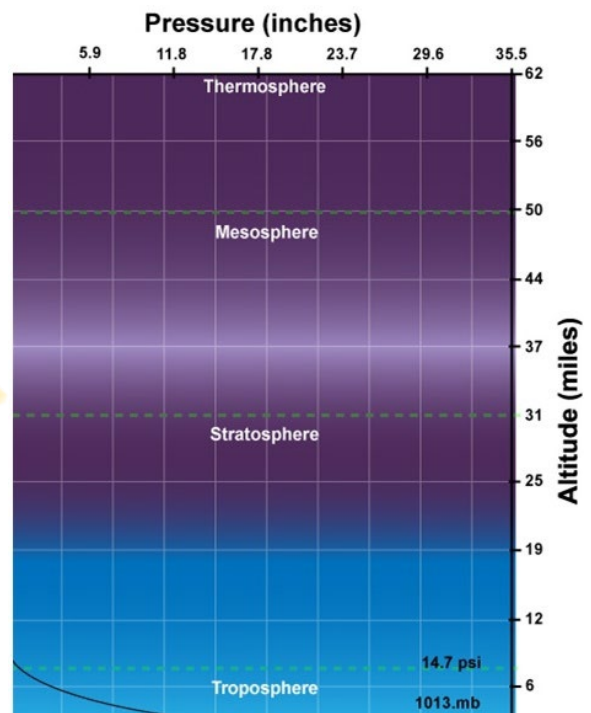
A body of air as deep as the atmosphere has tremendous weight.

- The weight of the atmosphere on an average person is about 20 tons

Pressure is the result of the weight of the air above the measurement position.

- The average pressure at sea level is 14.7 pounds per square inch (psi), which corresponds with 29.92 inches of mercury

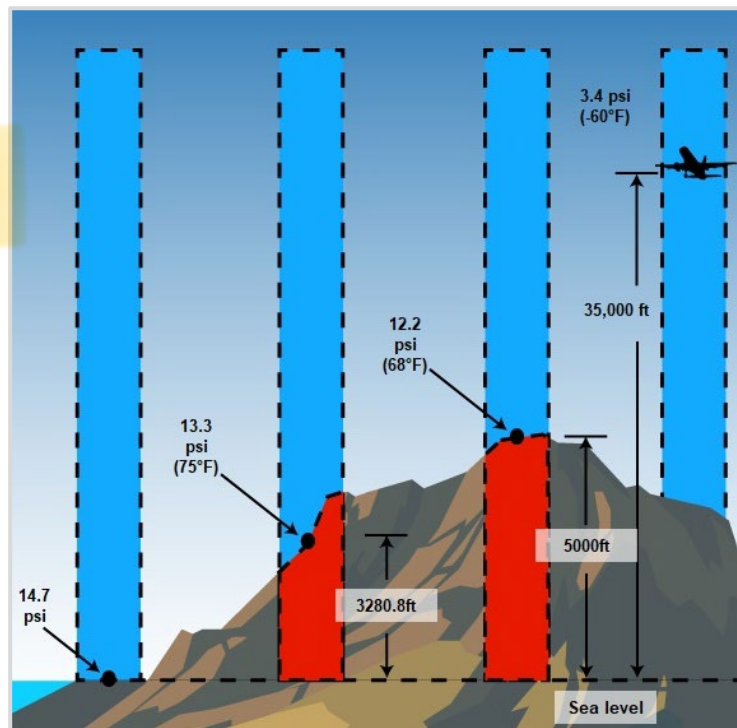
Note: In the atmosphere, both temperature and pressure decrease with altitude and have conflicting effects upon density. However, the fairly rapid drop in pressure as altitude is increased usually has the dominant effect. Hence, pilots can expect the total air density to decrease with altitude



Cold ↓ ↑ Pressure

Pressure decreases with height.

- At higher altitudes, there is less air above the measurement position and less weight
- For example, pressure decreases from 14.7 psi at sea level to 12.2 psi at 5,000 feet above sea level and to 3.4 psi at 35,000 feet (FL 350)
- Lower pressure results in less dense air



Water Vapor/Humidity

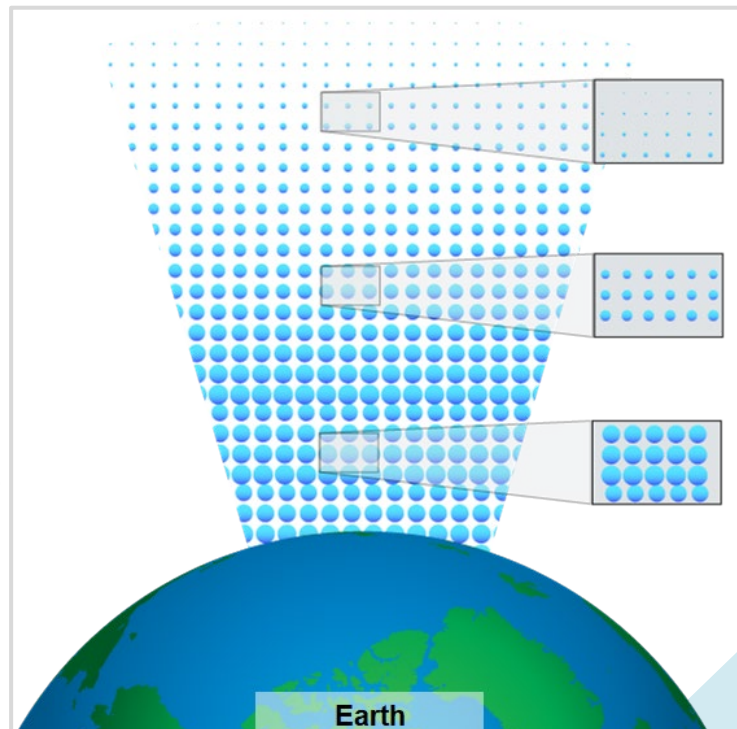
Moisture in the atmosphere is the invisible gas called water vapor.

- The higher the temperature, the greater amount of water vapor the air can hold
- Water vapor is lighter than air; consequently, moist air is lighter than dry air
- An increase in water vapor (higher humidity) results in a decrease in air density

Density and Density Altitude

Density is the mass of air per unit volume and is often described by the term density altitude.

- **Density altitude** is a term used to correlate aircraft performance, in a nonstandard atmosphere, to an altitude in the standard atmosphere corresponding to a particular value of air density
- Density altitude calculations are used by pilots to determine aircraft performance characteristics given the existing atmospheric conditions
- As the density of the air increases (lower density altitude), aircraft performance increases; conversely, as air density decreases (higher density altitude), aircraft performance decreases
- A decrease in air density means a higher density altitude; an increase in air density means a lower density altitude

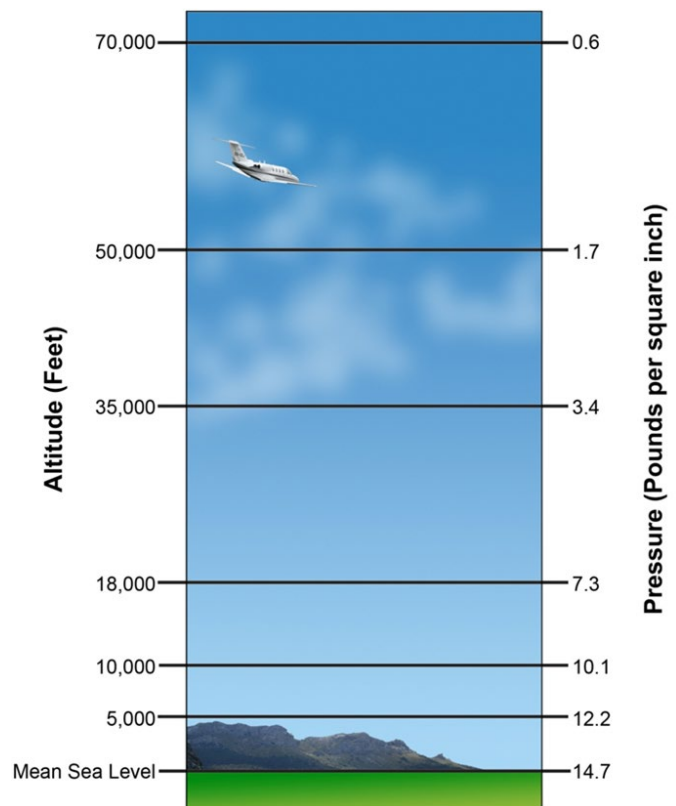


Density altitude is the vertical distance above sea level in the standard atmosphere at which a given air density is to be found.

Increased density altitude, such as in mountainous and high terrain areas with warm and humid air, can greatly reduce aircraft performance, including:

- Longer takeoff roll
- Longer landing roll
- Slower climb rate
- Reduced engine power output
- Landing speed increased

Note: The effects of high-density altitude on aircraft performance are going to greatly impact controller workload, e.g. *more spacing required, slower climbs, more time to clear the runway, etc.*



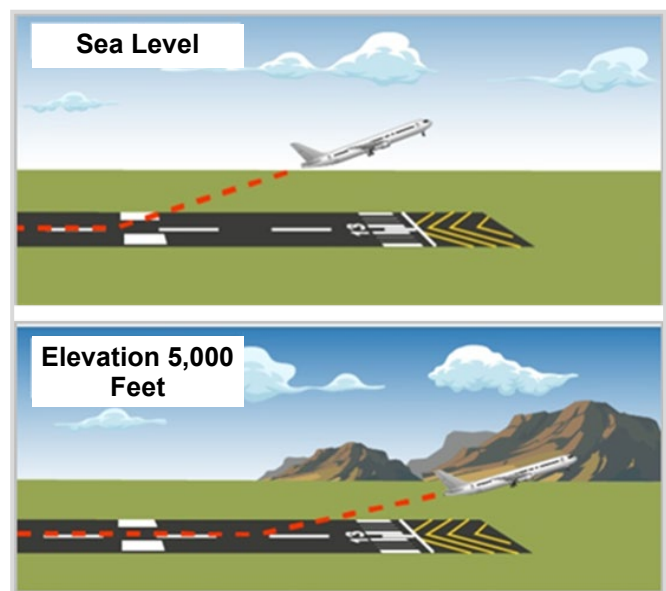
Effects of Atmosphere on Aircraft Performance

Effects of Altitude on Performance

An increase in altitude decreases atmospheric pressure and increases density altitude, which has a pronounced negative effect on flight. At higher elevation airfields:

- The length of the runway needed for takeoff roll will be increased
- The climb performance of an aircraft will be diminished
- The length of the runway needed for the landing roll will be increased
- The amount of power an engine can produce will be decreased

↓ Density ↑ Density Altitude



Effect of Temperature on Performance

Atmospheric density varies with temperature.

- When the air is heated, it expands and, therefore, has less density, increasing the density altitude

On a hot day, as compared to a cold day:

- Takeoff roll will be longer
- Rate of climb will be slower
- Landing speed will be faster
- Engine power output will be decreased

Effect of Humidity on Performance

Water vapor (humidity) is lighter than air; consequently, humid air is lighter than dry air. Therefore, as the water content of the air increases, the air becomes less dense, increasing density altitude and decreasing performance.

Increased humidity (decreased air density) has a less pronounced effect on density altitude than altitude and temperature but can still have a pronounced effect on flight.

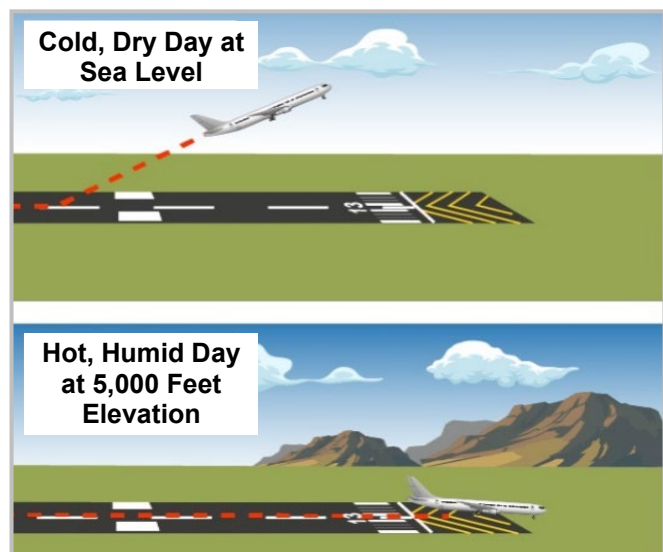
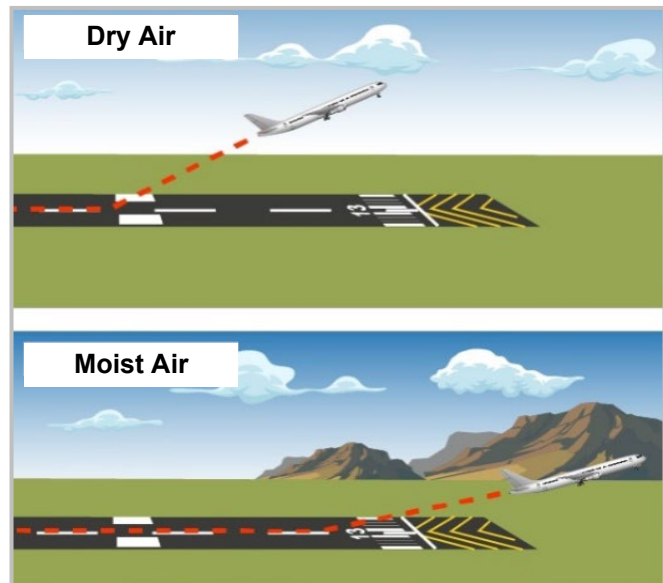
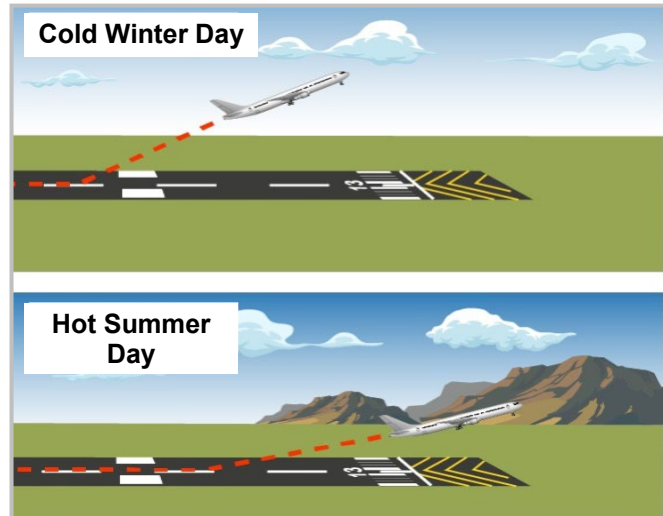
On a humid day, as compared to a dry day:

- Takeoff roll will be longer
- Rate of climb will be slower
- Landing speed will be faster
- Engine power output will be decreased

Combined Effects on Performance

High elevation airfields with hot and humid conditions will have very poor aircraft performance.

- Length of runway needed for takeoff roll will be increased
- Initial climb performance of an aircraft will be diminished
- Length of runway needed for landing roll will be increased
- Engine power output will be decreased





Knowledge Check D

REVIEW what you have learned so far about effects of atmosphere on aircraft performance. ANSWER the questions listed below.

1. The key properties of the atmosphere that affect air density and aircraft performance are: *(Select all correct answers that apply.)*
 - ☒ Altitude
 - ☐ Airfoils
 - ☒ Temperature
 - ☐ Turbulence
 - ☒ Humidity
2. What is the result of the weight of the air above the measurement position? *(Select the correct answer.)*
 - ☐ Density
 - ☒ Pressure
 - ☐ Precipitation
3. What are some of the ways increased density altitude can reduce aircraft performance? *(Select all correct answers that apply.)*
 - ☒ Longer takeoff roll
 - ☐ Shorter landing roll
 - ☒ Slower climb rate
 - ☒ Reduced engine power
 - ☒ Increased landing speed
4. How would temperature affect aircraft on a hot day? *(Select all correct answers that apply.)*
 - ☒ Longer takeoff roll
 - ☒ Faster landing speed
 - ☐ Increased engine power
 - ☒ Slower rate of climb
5. How would increased humidity affect aircraft performance? *(Select all correct answers that apply.)*
 - ☒ Longer takeoff roll
 - ☒ Faster landing speed
 - ☐ Increased engine power
 - ☒ Slower rate of climb

Effects of Atmosphere on Aircraft Performance Summary

Atmospheric conditions will make a difference in your job. On a warm day, you might direct air traffic differently than on a cold day. On a humid day, you might direct air traffic differently than on a dry day. Gaining a basic knowledge of how the atmosphere affects aircraft is essential to the safety of the pilots and passengers in your airspace.

PRIMARY AND SECONDARY FLIGHT CONTROLS

Purpose: The purpose of this lesson is to explain how flight controls work to control the movement of an aircraft and helicopter aerodynamics.

Objectives:

- Identify functions of primary and secondary flight controls and the movement around the aircraft axes
- Identify helicopter aerodynamics and controls

References for this lesson are as follows:

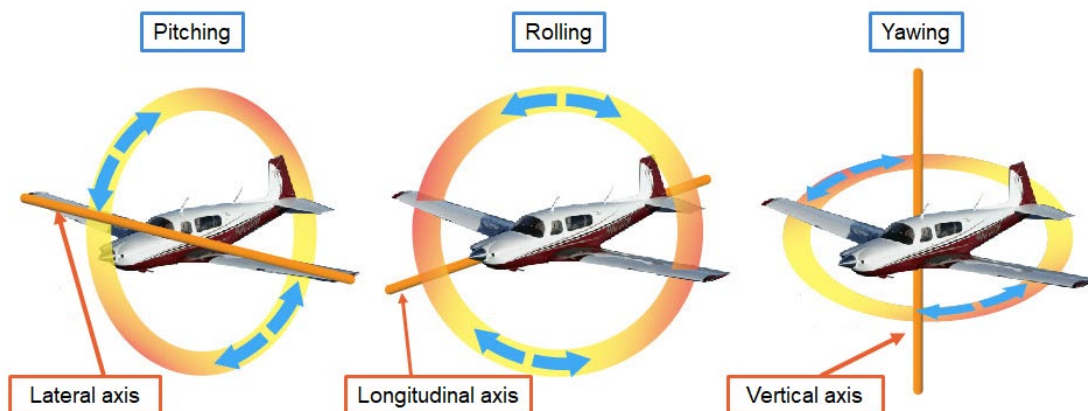
- FAA-H-8083-25, Pilot's Handbook of Aeronautical Knowledge
- FAA-H-8083-21, Rotorcraft Flying Handbook

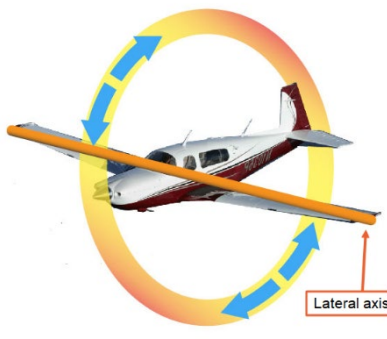
Rotational Axes of Aircraft

An axis is a straight line about which a body rotates. An aircraft has three axes of rotation.

- Longitudinal axis (roll)
- Lateral axis (pitch)
- Vertical axis (yaw)

The directions of rotation are always relative to the pilot view.



Longitudinal Axis (Roll)	The longitudinal axis is an imaginary straight line through the fuselage, nose to tail. ■ Movement around the longitudinal axis is called the roll movement ■ Controls angle of bank	 Longitudinal axis
Lateral Axis (Pitch)	The lateral axis is a line through the wing from wingtip to wingtip. ■ Movement around the lateral axis is called the pitch movement (nose up, nose down) ■ Controls angle of attack and aircraft pitch attitude	 Lateral axis
Vertical Axis (Yaw)	The vertical axis is a line through the center of gravity from top to bottom. ■ Movement around the vertical axis is called yaw movement ■ Controls left-to-right alignment of the longitudinal axis with respect to the relative wind ■ Controls the streamlined motion of the aircraft	 Vertical axis



Knowledge Check E

REVIEW what you have learned so far about primary and secondary flight controls. ANSWER the question listed below.

1. The three rotational axes on an aircraft are: (Select all correct answers that apply.)

- ☒ Roll
- ☐ Trim
- ☒ Pitch
- ☒ Yaw

Primary Control Surfaces

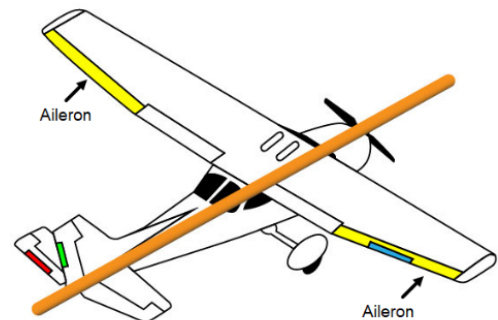
Control of the aircraft movement about its three axes of rotation is affected by the primary control surfaces.

- Ailerons (controls roll)
- Elevator (moves as a unit; controls pitch)
- Rudder (controls yaw)



Ailerons

Ailerons are hinged surfaces normally mounted on the outboard trailing edge of the wings. The ailerons rotate the aircraft around the longitudinal axis.



Movement of Ailerons

Left and right ailerons move simultaneously but in opposite directions.

- Lift increases on the down aileron, decreases on the up aileron

Moving ailerons induces adverse yaw.

- Adverse yaw is the tendency of the nose of the aircraft to yaw in the opposite direction of the turn
- Adverse yaw is caused by the drag of the “down” aileron

The control yoke controls the ailerons. The control yoke turns left or right, rotating the aircraft in whichever direction it is turned.

Normal state	 The top image shows a blue and white aircraft from a top-down perspective, flying straight. The bottom image shows a black control yoke with its grips in a neutral, upright position.
Left rotation	 The top image shows the aircraft banking to the left, with its wings tilted upwards on the left side. The bottom image shows the control yoke tilted to the left, with the left grip higher than the right.
Right rotation	 The top image shows the aircraft banking to the right, with its wings tilted upwards on the right side. The bottom image shows the control yoke tilted to the right, with the right grip higher than the left.

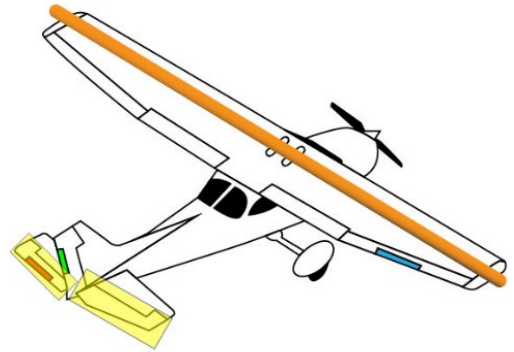
Elevator

The elevator is a hinged surface normally located on the rear of the horizontal stabilizer.

The elevator rotates the aircraft around the lateral axis.

- The elevator controls the pitch and angle of attack of the aircraft
- On some aircraft, the entire horizontal tail surface moves; this is also known as a **stabilator**

Note: The stabilator is a single-piece horizontal tail surface on an airplane that pivots around a central hinge point. A stabilator serves the purposes of both the horizontal stabilizer and the elevators.



Movement of Elevator

The control yoke moves forward and backward to control the elevator.

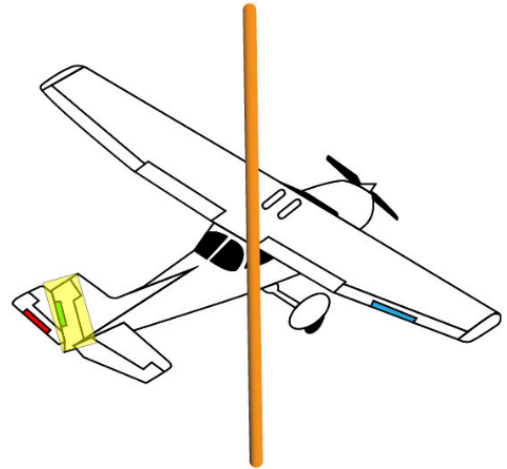
- Yoke pulled back, elevator would be up, nose up
- Yoke pushed forward, elevator would be down, nose down



Rudder

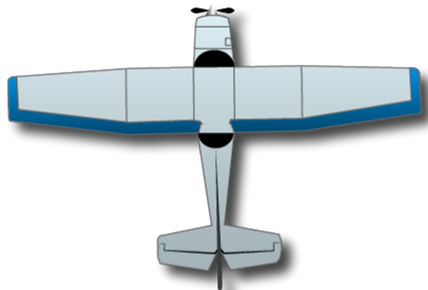
The rudder is in the aft of the vertical stabilizer.

The rudder rotates the aircraft around the vertical axis. The rudder controls the yaw of the aircraft.



Movement of Rudder

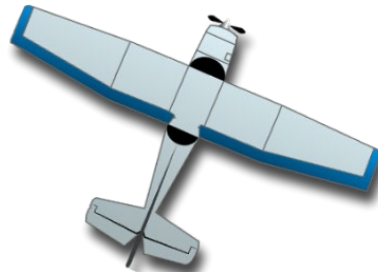
The rudder is controlled by rudder pedals.



Depress left pedal, aircraft will yaw left



Depress right pedal, aircraft will yaw right





Knowledge Check F

REVIEW what you have learned so far about primary and secondary flight controls. ANSWER the questions listed below.

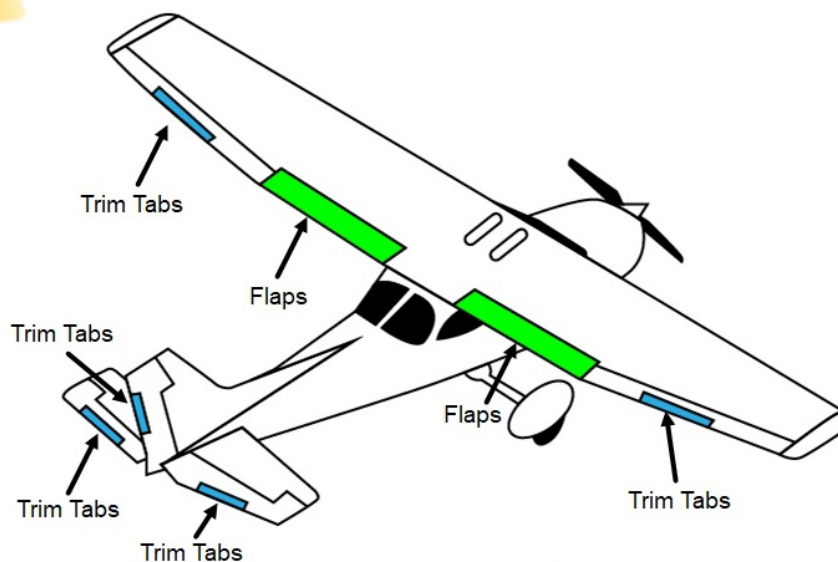
1. Which two movements are controlled by the control yoke? (Select the correct answer.)
 - ☐ Pitch and yaw
 - ☒ Roll and pitch
 - ☐ Yaw and roll
2. Identify the primary control surfaces with their locations on the aircraft. Write the answer on the correct line.



Secondary Control Surfaces

Secondary control surfaces include:

- Trim tabs
- Flaps



Trim Tabs

Trim tabs are small, adjustable, hinged surfaces on the trailing edge of the primary control surfaces. The purpose of trim tabs is to **lessen the manual pressure** the pilot must apply to the control surfaces.

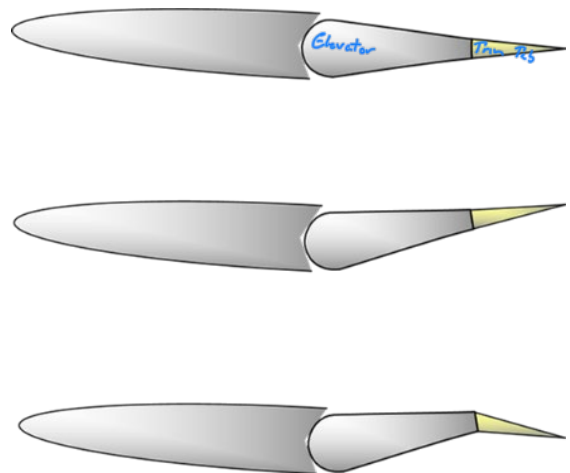
- Aileron trim tabs are generally used on large aircraft
- Elevator and rudder trim tabs are common on all aircraft



Elevators and Trim Tabs

Trim tabs hold the control surface in position aerodynamically.

- Control surface can still be moved by pilot
- Relieves pressure on controls



Trim Tab Control

Trim tab controls are either **manual or electric**.

Note: Some light airplanes have trim tabs that are ground adjustable.



Could be an Emergency Situation

Flaps

Flaps are located inboard on the wing's trailing edge and are used to increase lift. Flaps are extended in increments described as degrees from the full up position of 0 degrees.

- Usually, three or four positions can be selected, i.e., 10, 20, 30, 40 degrees
- Flaps extend on both wings at the same time

The extension of the flaps increases the camber and, on some types, increases the wing area.

- Increases lift
- Increases drag
- Lowers stall speed
- Allows steeper approach to runway without increased speed

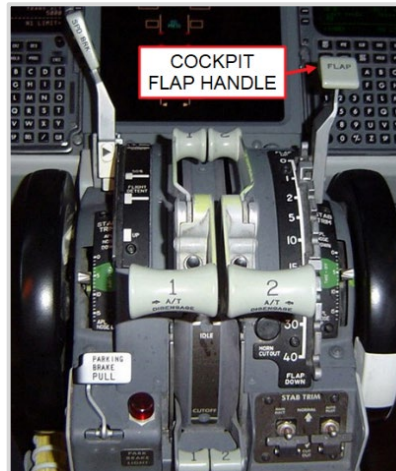


Flaps are mostly used for takeoff and landing.

Cockpit Flap Handle

Flaps are adjusted:

- Manually
- Electrically
- Hydraulically



Knowledge Check G

REVIEW what you have learned so far about primary and secondary flight controls. ANSWER the questions listed below.

1. Which of the following is a primary control surface? (Select the correct answer.)
 - ☐ Variable pitch propeller
 - ☐ Flap
 - ☒ Rudder
2. The extension of flaps causes an increase in _____. (Select the correct answer.)
 - ☐ Stall speed
 - ☐ Airspeed
 - ☒ Drag

Basic Helicopter Aerodynamics

The forces acting on helicopters are the same as those acting on fixed-wing aircraft.

- Lift
- Thrust
- Weight
- Drag

Lift is provided by rotor blades.

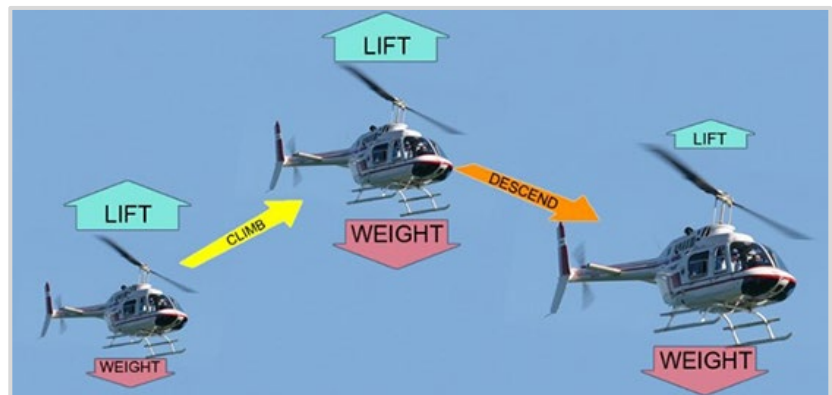
- Each blade is shaped like an airfoil
 - Bernoulli's Principle applies
- The rotor blades are moved through the air by the engine; when the blades are in motion, they act as a wing



Relationship of Lift and Weight in a Hover

A combination of Revolutions Per Minute (RPMs) and blade pitch controls:

- Vertical ascent
 - Lift is greater than weight
- Hovering
 - Motionless flight over a reference point
 - Constant heading and altitude
 - Thrust and lift equals weight and drag
- Vertical descent
 - Weight is greater than lift



Helicopter Controls

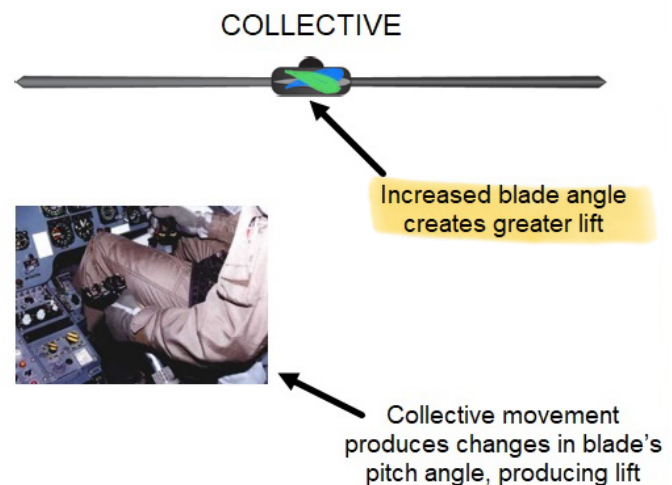
Throttle

The throttle is mounted on the forward end of the collective pitch lever in the form of a motorcycle type twist grip. The function of the throttle is to regulate the RPMs.

Collective *CO P*

The collective controls the pitch of the rotor blade (angle of attack). The greater the blade angle, the greater the lift produced.

Note: It is important to note that communications with helicopters in flight are limited at times because the pilot has both hands on the controls.



Cyclic

The **cyclic** controls the tilt of the rotor blade, which controls the direction of flight.

The cyclic is pushed in the direction that the helicopter is to be moved.

- The tilt of the rotor blades creates thrust in the direction of movement



Aut rotation

Aut rotation is the state of flight where the main rotor system is being turned by the action of relative wind rather than engine power.

- Allows the aircraft to make a controlled landing when the engine is no longer providing power

A helicopter transmission is designed to allow the main rotor to rotate freely in its original direction if the engine stops.

- At the instant of engine failure, the blades will be producing lift and thrust as a result of their angle of attack and velocity
- As the helicopter descends, the upward flow of air provides sufficient thrust to maintain rotor RPMs and lift throughout the descent



Video – Aut rotation (1:08 mins.)



Knowledge Check H

REVIEW what you have learned so far about primary and secondary flight controls. ANSWER the question listed below.

1. Helicopter controls include: *(Select all correct answers that apply.)*
 - Throttle
 - Collective
 - ☐ Aileron
 - Cyclic

Primary and Secondary Flight Controls Summary

Gaining a basic knowledge of aircraft flight controls and aerodynamics will help you communicate accurately and professionally with pilots concerning their aircraft. This knowledge could be invaluable to you in your role as an air traffic control specialist in the case of an aircraft emergency.

HAZARDS AFFECTING FLIGHT

Purpose: The purpose of this lesson is to explain different conditions and situations that create hazards that affect flight.

Objective:

- Describe hazards that affect flight

References for this lesson are as follows:

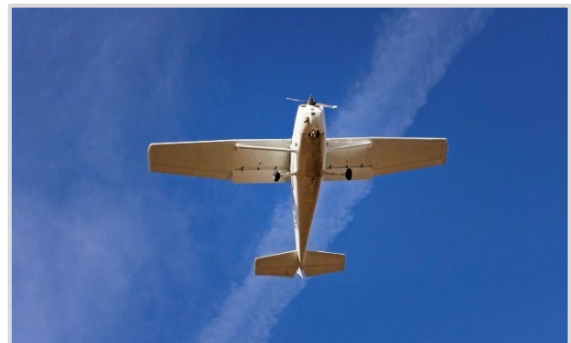
- FAA-H-8083-28, Aviation Weather Handbook
- FAA-H-8083-25, Pilot's Handbook of Aeronautical Knowledge
- Aeronautical Information Manual (AIM)

Stalls



Stalls are the most common cause of light aircraft accidents. A stall occurs when the airfoil (wing) exceeds the "critical angle of attack," which is approximately 15 to 20 degrees on most airfoils.

- During flight, the airstream remains attached to the wing surface and lift is produced when operating within the normal angle of attack range
- As the critical angle of attack is approached, the smooth flow of the airstream begins separating from the rear of the upper wing surface and ceases to produce lift
- As the critical angle of attack is exceeded, the separation of smooth flow moves forward to the area of the highest camber and creates a wing stall

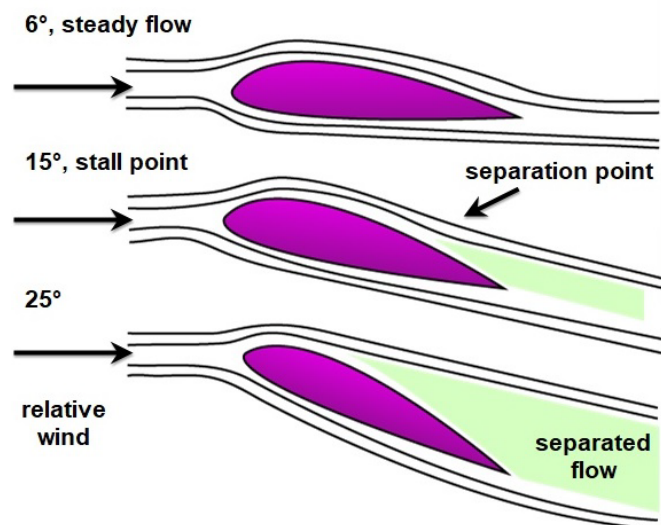


Causes of Stalls

The three primary causes of stalls are:

- Insufficient airspeed
- Excessively violent flight maneuvers
- Severe wind shear

Video – Stalls – Wind Tunnel (1:14 mins.)



Icing

The three primary types of icing are:

- Structural icing
- Pitot-static system icing
- Carburetor icing

Structural Icing

Structural icing changes the shape of the airfoil.

- The greatest hazard of this type of icing is airfoil distortion, which disrupts smooth airflow reducing lift and also adds weight
- Many IFR-equipped aircraft have anti-icing and/or deicing equipment



Pitot-Static System Icing

When pitot tube icing occurs, the airspeed indicator becomes unreliable.

- Pitot heat is used on many aircraft to prevent icing

Although rare, when static vent icing occurs, all three instruments are affected, e.g., airspeed, vertical speed indicator, and altimeter.

- An alternate static air vent, although not as accurate, is installed inside the cabin on some aircraft

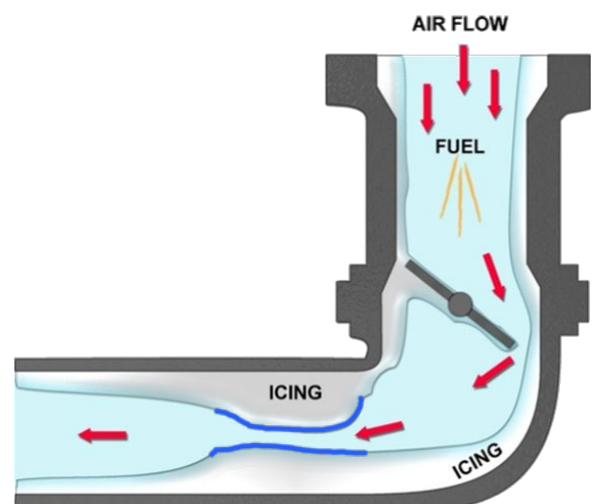
*Pitot Tube
- Tube on outside of aircraft
that reads airspeed*



Carburetor Icing

Carburetor icing reduces the fuel/air flow to the engine.

- This can cause complete engine failure by starving the engine of fuel and air
- Occurs most often between 20-70° F under conditions of high humidity
- Lowered pressure and vaporization in the carburetor lowers the temperature of the fuel/air mixture to the point where any water vapor or moisture present will freeze, forming ice or frost inside the carburetor
- Carburetor ice can be cleared by adding carburetor heat, which recirculates heated air from engine, but this is primarily for anti-icing, not de-icing



Aircraft System Failures

Aircraft system failures may occur due to:

- Electrical failures
- Mechanical failures
- Hydraulic failure
- Engine failure
- Engine fire

When you become aware of an unusual situation, use all available resources to assist the aircraft and notify your supervisor.



Electrical Failure

During partial electrical failure, some instruments and systems are affected.

When complete electrical failure occurs, there is a loss of:

- Some instruments
- Flaps on some aircraft
- Radios and navigation and transponder equipment
- Lights

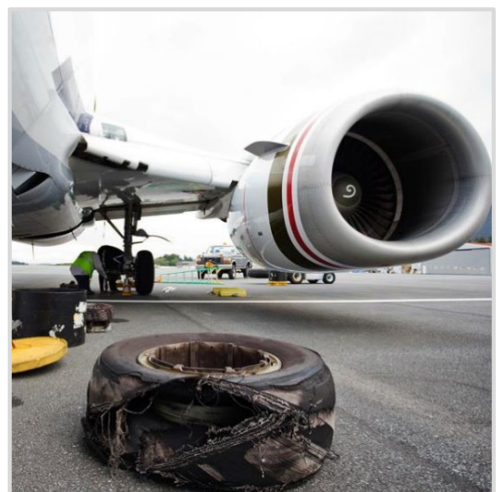
Air traffic controllers should be aware of these losses and assist in any way possible, e.g., priority handling, clear conflicting traffic, alert emergency equipment.



Mechanical Failure

Mechanical failures include:

- Landing gear
- Blown tire
- Wheel off
- Panel off
- Flight controls
- Windshield



Hydraulic Failure

Hydraulic failure affects landing gear, flaps, and brakes on some aircraft.

Handling a hydraulic failure may require long runways, emergency equipment, etc.



Engine Failure

Engine failure may affect:

- Engine-driven vacuum system for instruments
- Hydraulic power
- Electrical power
- Pressurization

Engine failure may result in:

- Loss of altitude
- Forced landing

Air traffic controllers should assist pilots by advising them of the nearest airport suitable for landing.



Engine Fire

An engine fire is usually controllable.

- The indication to the pilot of an engine fire is via the fire warning light

A cabin/cockpit fire is extremely serious.

Specialists who are advised by pilots of a fire warning light or a cockpit/cabin fire may expect either a bailout or a request for immediate landing.





Knowledge Check I

REVIEW what you have learned so far about hazards affecting flight. ANSWER the questions listed below.

1. The most common cause of light aircraft accidents is: *(Select the correct answer.)*
 - ☐ Icing
 - ☒ Stalls
 - ☐ Mechanical failures
 - ☐ Engine fires

2. Pitot tube icing causes which of these instruments to become unreliable? *(Select the correct answer.)*
 - ☐ Altimeter
 - ☐ Vertical speed indicator
 - ☒ Airspeed indicator
 - ☐ Fuel level indicator

Hazards Affecting Flight Summary

Stalls, ice, and system failures are all hazards that can happen to an aircraft, without warning, at any time. Gaining a basic understanding of these hazards will better prepare you in determining how best to assist the aircraft for a successful flight and safe landing.

SUMMARY

The purpose of this module was to provide basic aeronautical information that will help you communicate with pilots concerning the operation of their aircraft.

In accordance with FAA-H-8083-25, Pilot's Handbook of Aeronautical Knowledge; the Aeronautical Information Manual (AIM); FAA-H-8083-21, Rotorcraft Flying Handbook; and FAA-H-8083-28, Aviation Weather Handbook; you should now be able to:

- Identify primary and secondary sources of lift
- Identify types and parts of airfoils
- Identify forces affecting flight, their interrelationships, and their effects on aircraft performance
- Identify effects of altitude, temperature, and humidity on aircraft performance
- Identify functions of primary and secondary flight controls and the movement around the aircraft axes
- Identify helicopter aerodynamics and controls
- Describe hazards that affect flight

KNOWLEDGE CHECK ANSWERS

Knowledge Check A

1. Pressure; 2. Bernoulli's Principle

Knowledge Check B

1. Camber; 2. Wing, Horizontal tail surfaces, Propeller

Knowledge Check C

1. b, c, d, a

Knowledge Check D

1. Altitude, Temperature, Humidity; 2. Pressure; 3. Longer takeoff roll, Slower climb rate, Reduced engine power, Increased landing speed; 4. Longer takeoff roll, Faster landing speed, Slower rate of climb; 5. Longer takeoff roll, Faster landing speed, Slower rate of climb

Knowledge Check E

1. Roll, Pitch, Yaw

Knowledge Check F

1. Roll and pitch; 2. Rudder, Elevator, Aileron

Knowledge Check G

1. Rudder; 2. Drag

Knowledge Check H

1. Throttle, Collective, Cyclic

Knowledge Check I

1. Stalls; 2. Airspeed indicator